

Project Classification: Blockchain & Web3 RWA Platform

Executive Architectural Overview & Supervisor Defense Brief

Is this a Blockchain Project?

YES, ABSOLUTELY. This system is classified as a **Hybrid Web3 Real-World Asset (RWA) & Escrow Platform**. The core custody of investor funds, investment agreements, unit allocation, and harvest profit-sharing are fully decentralized and enforced on-chain via EVM Smart Contracts (written in Solidity 0.8.x), connected to a real-time reactive event ingestion layer.

1. Key Industry Terminology for Presentation

"Hybrid Web2/Web3 Real-World Asset (RWA) & Escrow Platform"

Just like modern Web3 applications (OpenSea, Aave, Uniswap), this platform combines a high-speed web application for onboarding and document indexing with immutable, decentralized smart contracts for all monetary value flows, escrow custody, and investor rights.

2. Concrete Evidence: The 4 Core Blockchain Pillars

1. Smart Contracts Layer (Solidity)

Financial rules and escrow are executed on-chain via 6 contracts in `contracts/src/` :

- `Escrow.sol` : Trustless fund lockup & milestone payouts.
- `DealContract.sol` & `DealFactory.sol` : On-chain deal creation & investor unit allocation.
- `ProfitDistribution.sol` : Code-enforced harvest profit splits.

2. Web3 Wallet Layer (Frontend)

Integrated with **Wagmi**, **Viem**, and **RainbowKit**:

- Supports MetaMask, Coinbase Wallet, and WalletConnect.
- Users transact with cryptographic Ethereum addresses (`0x...`).
- Deal commitments are signed cryptographically by investors.

3. Live Event Listener (Backend)

Reactive blockchain telemetry in `apps/api/src/blockchain/` :

- Continuous indexing using **Viem** public client.
- Handlers: `EscrowHandler`, `InvestmentHandler`, `ProfitHandler`.
- Automatically triggers DB updates upon block confirmation.

4. On-Chain Reconciliation Engine

Continuous audit loop in `reconciliation.service.ts` :

- Audits PostgreSQL accounting records against live EVM state.
- Detects balance drift between database ledgers and blockchain escrow.
- Ensures zero discrepancy between off-chain cache & on-chain truth.

3. Why Use Blockchain Here? (Business & Technical Value)

- **Trustless Milestone Escrow:** International or retail investors do not need to trust an unvetted local bank or farmer. Capital is held in a smart contract and released only when verification milestones are met.
- **Code-Guaranteed Profit Sharing:** Pre-agreed percentage splits (e.g. 70% farmer / 30% investors) execute programmatically without manual intermediaries.
- **Frictionless Cross-Border Liquidity:** Enables global capital to fund agriculture without multi-day international wire transfers or currency conversion loss.

- **Immutable Audit Trail:** Every funding, release, and dividend distribution is recorded immutably on the public EVM ledger for full transparency.

4. Quick Supervisor Q&A Reference Table

Supervisor Question	Direct Technical Answer
Is this genuinely a blockchain project?	Yes. It is an RWA (Real-World Asset) DeFi & Escrow application with on-chain settlement.
Where does the blockchain live?	In contracts/src/ (Solidity 0.8.x), deploying to Hardhat, Base Sepolia, or Ethereum.
What happens on-chain?	Deal deployment, capital escrowing, investor unit issuance, and dividend distribution.
How does the web app talk to it?	Via Web3 wallets (Wagmi/RainbowKit) on frontend and Viem event listeners on backend.
Why not just use SQL?	SQL cannot provide trustless escrow, code-enforced dividends, or global permissionless liquidity.